



A comparison of endogenous and exogenous group formation to increase cooperation[☆]

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ARTICLE INFO

JEL Classification:

C92

D02

D71

Keywords:

Cooperation

Group formation

Decision rights

Democracy premium

ABSTRACT

Grouping cooperatively disposed subjects with one another has been shown to slow the decline of contributions in public goods games. So far, similar findings have been obtained both when groups are exogenously formed by the experimenter and when they are endogenously formed by algorithms taking subjects' preferences for partners into account. We designed an experiment in which internally homogeneous groups – composed of like-minded subjects – can form either exogenously or endogenously, testing the conjecture that endogenously formed groups of cooperators will outperform similar exogenously formed groups due to subjects' decision rights in the endogenous procedure. We find that the conjecture is not supported, discussing potential explanations.

1. Introduction

Overwhelming evidence from laboratory and field experiments indicates that unrelated individuals tend to cooperate despite individual costs (Bowles and Gintis 2011; Fehr and Gächter, 2000; Raihani 2021; Rand and Nowak, 2013). The extent to which individuals cooperate has been under intense scrutiny and a frequent observation has been that cooperation tends to decline over time (Gächter and Herrmann, 2009). The explanation put forward in the behavioral economics literature is that most individuals are conditional cooperators and tend to reduce their contribution in response to others' free-riding (Gächter, 2014). Different solutions have been tested with the objective to reduce the decline in cooperation over time, such as punishment (Fehr and Gächter 2000; Masclet et al., 2003), communication (Balliet, 2010; Bochet et al., 2006; Ostrom et al., 1992), and group sorting (Gächter and Thöni, 2005; Guido et al., 2019; Page et al., 2005). Chaudhuri (2011) notes that assortative group formation is one of the most effective non-punitive solutions to sustain cooperation. Despite the growing literature investigating group formation mechanisms, it remains unclear whether forming groups of like-minded individuals endogenously based on individual preferences leads to higher cooperation than if such groups are

created exogenously by the experimenter. There is empirical evidence that people value procedural autonomy (Deci and Ryan, 1985), and that there is an intrinsic value to decision rights (Bartling et al., 2014). However, it is unclear whether allowing subjects to choose their group mates can increase cooperation compared to an exogenous mechanism. Our aim in this study is to fill this gap by providing experimental evidence regarding the effect of endogenous and exogenous group formation mechanisms on cooperation, holding the type composition of groups constant.

We design a public goods experiment with two phases. In the first phase, subjects play a one-shot game in groups of 3. The one-shot public goods game provides a measure of people's intrinsic disposition to cooperate that is not influenced by reputational concerns (Gächter and Thöni, 2005). Subjects receive no information about their earnings nor about the decisions of the others in the same group after phase 1. The second phase starts with a regrouping stage where new groups of 3 are either created based on subjects' own preferences elicited in a ranking (our endogenous mechanism) or by the experimenter (our exogenous mechanism). To rank others, subjects are given the individual contribution from phase 1 of each other subject in the session. Subjects are asked to indicate their most preferred partners knowing that they will

[☆] The experimental protocol (Protocol #3598) has been certified as exempt from IRB supervision by the Claremont Graduate University's Institutional Review Board on February 6, 2020. Link to the pre-registration document: <https://aspredicted.org/zi95w.pdf>.

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<https://doi.org/10.1016/j.socex.2025.102451>

Received 15 July 2024; Received in revised form 22 September 2025; Accepted 23 September 2025

Available online 25 September 2025

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play a 10 round public goods game. As in [Page et al. \(2005\)](#), our ranking led to perfectly sorted groups in the order of their level of cooperation in the first phase of the experiment. Indeed, in the endogenous condition, groups are as homogeneous (composed of like-minded subjects) as in the exogenous grouping described next.

Our exogenous mechanism is similar to [Gächter and Thöni's \(2005\)](#) where the three highest contributors from phase 1 are grouped together, then the next three, until the lowest three contributors end up in the same group. By design, the exogenous mechanism leads to the creation of homogeneous groups, thus making it possible to compare the exogenous condition to our endogenous mechanism. It is worth noting that in all sessions, subjects go through the ranking stage at the beginning of phase 2. However, subjects are informed that new groups will be created either based on their preferences expressed in the ranking or based on a rule predefined by the experimenter. Comparing cooperation levels in the endogenously formed homogeneous groups with that in the exogenously formed homogeneous groups allows us to identify the causal effect of the group formation procedure on contributions to a public good. Furthermore, our design allows us to study heterogeneity effects, that is the impact of the group formation procedure on individuals with high, medium, and low intrinsic dispositions to cooperate. In particular, irrespective of whether individuals are sorted endogenously or exogenously in our experiment, high types always get their most preferred partners. Thus, in the case of high types, differences in behavior, if any, cannot be attributed to factors other than the method employed to create groups.

We find that subjects cooperate more in exogenously formed homogeneous groups than when such groups are endogenously created. The difference between the two is mainly driven by the contribution behaviors of subjects with a high disposition to cooperate as measured by these individuals' contributions in the phase 1 one-shot game. Our experiment allows us to eliminate different explanations for this result and to formulate two conjectures. The first is that the ranking outcome in the endogenous condition may raise the salience of the view that others are self-interested. Indeed, we find that subjects give better ranking to higher contributors and they may conclude that the reason for that is primarily material self-interest. The second is that good ranks received by high contributors may raise the salience that one has high moral credits, which may provide a license for subsequent bad deeds. Since our experimental findings were largely unanticipated by us and the potential explanations were thus formulated after the fact, we are unable to perform direct tests of these conjectures and must leave it to others to find ways of generating data that might allow direct tests of these hypotheses. We also cannot rule out the possibility that this is a one-off result, comparable to the early finding of [Andreoni \(1988\)](#) that stranger groups contribute more than partner groups, which subsequent studies did not replicate ([Andreoni and Croson, 2008](#)).

Our research contributes to two branches of the behavioral economics literature. We contribute, first, to the experimental literature on the effect of group sorting on cooperation in social dilemma games. [Guido et al. \(2019\)](#) surveyed more than 60 papers that implemented either an endogenous or an exogenous sorting mechanism. Their survey and meta-analysis provide suggestive evidence that there is no difference in cooperation levels between experiments that implemented an endogenous sorting and studies that used an exogenous mechanism to create homogeneous groups. Our study is the first to compare the two sorting mechanisms using the same experimental environment. Furthermore, we are able to identify the behaviors of different types of subjects (high, medium, low cooperators) in exogenously and endogenously formed groups, which has not been investigated so far. The other branch of the behavioral economics literature to which we contribute investigates the value of procedural autonomy and decision rights ([Bartling et al., 2014](#); [Deci and Ryan, 1985](#); [Frey and Jegen, 2001](#); [Frey et al., 2004](#)). Given that groups are formed based on subjects' expressed preferences in the endogenous condition, while this is not the case with the exogenous procedure, subjects might perceive more autonomy in the

former environment, which may affect their contributions.¹

In the next section, we review the related literature. [Section 3](#) presents the experimental design. [Section 4](#) discusses our predictions. [Section 5](#) presents the results. [Section 6](#) discusses the main results in light of the previous literature. [Section 7](#) concludes.

2. Literature review

Two branches of the literature in behavioral economics are closely related to the present study: (a) the literature on endogenous and exogenous group formation in the context of social dilemma games, and (b) the literature on procedural autonomy and the intrinsic value of decision rights.

2.1. Endogenous and exogenous group formation

[Guido et al. \(2019\)](#) offer an exhaustive review of the group formation literature with a specific focus on endogenous and exogenous sorting mechanisms and their respective impact on cooperation. Their survey identified 62 research items (published and unpublished) that implemented one of the three types of sorting:

1. Endogenous sorting where subjects playing a social dilemma game have some ability to form or leave groups on their own accord;
2. Endogenous sorting with an opt-out option where subjects can avoid the game entirely;
3. Exogenous sorting where group composition is determined by the experimenter with or without subjects' knowledge.

To illustrate the main experimental features of an experiment using an endogenous sorting and of an experiment using an exogenous sorting, we chose two highly cited papers: [Page et al. \(2005\)](#) and [Gächter and Thöni \(2005\)](#). In [Page et al. \(2005\)](#), each experimental session consisted of 16 subjects playing 20 rounds of a finitely repeated public goods game in groups of four. In the baseline, groups were formed randomly and group composition remained fixed for the duration of the experiment. The only difference between their baseline and the regrouping treatment is that in the latter the authors implemented a regrouping stage at the end of each third round. Each subject was shown a list of each of the other 15 subjects' average contribution over the experiment so far. Subjects were then offered the opportunity to express a preference among possible future partners by ranking them. In their experiment, ranking was costly: subjects were charged 25 experimental cents for the first ranking decision, and 5 experimental cents for each additional ranking. In the regrouping treatment, groups were therefore formed based on subjects' preferences.² [Page et al.](#) found that since most subjects preferred to group with higher contributors, the higher four contributors ended up in the same group, followed by near to highest, etc. Thus, the mechanism perfectly sorted the groups in the order of their members' contribution to the public good. The regrouping mechanism significantly increased contributions compared with the baseline, with most subjects continuing to contribute more in later periods the more

¹ Our study could also be related to the literature on the effects of endogenous and exogenous institutions on cooperation. [Tyran and Feld \(2006\)](#), [Dal Bo et al. \(2010\)](#), [Sutter et al. \(2010\)](#), and [Markussen et al. \(2014\)](#) found a democracy premium, i.e., a higher level of cooperation under democratically chosen institutions compared to when the same institutions are implemented exogenously. However, our study is not about institutional choices, e.g., we do not study whether subjects would vote for an endogenous or an exogenous group formation mechanism.

² Thus, total cost of assigning rank increased with the number of others ranked. Any subjects not assigned a rank were treated as sharing the median of the unutilized rank numbers. Details of how ranks were used to form groups are provided below when we discuss the closely related variant used in our own design.

they had contributed in earlier ones.

When it comes to exogenous sorting, [Gächter and Thöni \(2005\)](#) provided one of the first experimental analyses of its effect on cooperation. In their experiment, subjects first participated in a one-shot linear public goods game in randomly formed groups of three. Subjects were informed that they would play the game just once in that phase of the experiment, but that some other phase would follow. However, they were not given further details about the next phase. This was done in order to ensure that subjects' decisions in the one-shot game are not strategically biased and reflect their "true" cooperative preferences.³ At the end of the one-shot game, subjects received instructions for the second phase. The second phase consisted of a ten rounds linear public goods game with fixed group composition. Before playing the second phase, subjects were shown the individual contributions of their new group mates from the phase 1 one-shot game. Subjects were then informed about their new group members' contributions in the one-shot game. The authors also implemented a baseline, where after playing the one-shot game, groups were formed randomly and each subject was informed about their new group mates' contributions in the one-shot game. In [Gächter and Thöni's](#) experiment, the highest three contributors ended up in the same group, followed by the next to highest, etc. While the composition of groups seems very similar to the one achieved by [Page et al. \(2005\)](#), in [Gächter and Thöni](#) this happened by design. Their results also indicate that cooperation is higher in the sorted groups than in the randomly created groups, and that initially higher contributors continue to contribute more in later periods.

The two papers show that it is possible to create groups where like-minded individuals are grouped together and that this can be achieved endogenously or exogenously. Each of these sorting mechanisms has been investigated separately against a baseline with randomly created groups and the results show that cooperation is higher in sorted groups than in randomly created ones, with cooperation rates varying in a persistent pattern across groups. However, it is not clear whether endogenously created groups cooperate more than exogenously formed ones, controlling for the type composition of groups (that is, subjects' cooperative types in each group based on their contributions in the first round of the public goods game: high, medium, or low contributors). Our paper provides the first experimental evidence on this question.

2.2. The intrinsic value of decision rights

The second branch of the literature in behavioral economics that is related to this paper studies the intrinsic value of decision rights. There is evidence from psychology and economics that autonomy is an essential ingredient of human motivation ([Deci and Ryan, 1985](#); [Festé and Garrouste, 2015](#); [Frey and Jegen, 2001](#); [Tirole, 2009](#)). One potential mechanism through which autonomy may affect cooperation is procedural utility – the idea that people derive satisfaction not only from outcomes but also from the fairness or meaningfulness of the process leading to those outcomes ([Frey et al., 2004](#)). In our context, giving subjects a voice in choosing their group mates may increase their sense of ownership over group composition and outcomes, spurring their desire to successfully cooperate with fellow group members.

From an experimental perspective, [Bartling et al. \(2014\)](#) developed a method that allowed them to provide clean evidence for the existence of an intrinsic value of decision rights beyond their instrumental benefits. In their experiment, subjects acted as principals deciding whether to retain decision-making authority or delegate it to an agent. The study isolated the intrinsic value subjects placed on holding decision rights, beyond any material outcomes. [Bartling et al.](#) discuss the potential

implications from their results for corporate governance, human resource management, and optimal job design. One important consequence from showing that people value decision rights may be that forming groups based on people's decisions will result in higher cooperation than when the same groups are formed exogenously. Indeed, there may be a procedural utility from knowing that groups are formed based on people's choices, which in turn could make them contribute more to their group's public good.

3. Experimental design

Our experiment consists of 18 sessions involving 270 subjects. In each experimental session, 15 inexperienced subjects from the Burgundy School of Business (Dijon) played public goods games in groups of three. All sessions were conducted in person at the Laboratory for Experimentation in Social Sciences and Behavioral Analysis (LESSAC) between September 2021 and October 2021.⁴

The experiment is composed of two phases. Subjects play the first phase not knowing what is coming next. We discuss the first phase of the experiment because it is the same across conditions. We then present the second phase of the experiment that has elements that are specific to each of our three conditions: endogenous condition (ENDO), exogenous homogenous condition (EXO—H), exogenous mixed condition (EXO—M). It is worth noting that our ENDO condition resembles the endogenous sorting mechanism used by [Page et al. \(2005\)](#), while our EXO—H condition parallels the exogenous sorting mechanism employed by [Gächter and Thöni \(2005\)](#). By comparing the two, we aim to bridge the gap between these influential contributions to the literature on human cooperation.

3.1. Phase 1 of the experiment

In the first phase of the experiment, groups of three are formed randomly. Each subject has 20 tokens that they can allocate to two accounts: an individual account and a group account. Each token allocated to the subject's individual account yields 1 experimental currency unit (ECU), while a token allocated to the group account by any subject in the group yields 0.6 ECU to each group member. Thus, every token placed in the group account yields 1.8 ECU to the group. The dominant strategy for a selfish money-maximizing individual is to allocate all 20 tokens to their private account. However, group earnings are maximized when all subjects allocate their entire endowment to the group account. We further refer in the paper to allocations to the group account as "contributions to the public good" or "cooperation," as is common in the public goods literature (e.g., [Andreoni, 1995](#); see [Chaudhuri, 2011](#), for a survey of the literature).

Subjects are informed that the information about how much each group member allocated to the group account as well as the subject's own individual earnings will only be provided at the very end of the experiment. Thus, subjects receive no feedback after the first phase of the experiment. Furthermore, the instructions indicate that there is only one decision to be made in the first phase. We use similar wording as in [Gächter and Thöni \(2005\)](#) to ensure that subjects perceive the game they play as a one-shot game.

3.2. Phase 2 of the experiment

In the second phase of the experiment, subjects receive new instructions informing them that the new phase is similar to the first one: each subject receives 20 tokens and needs to decide how they wish to allocate their endowment between two accounts. The group size is the

³ However, as noted by [Gächter et al. \(2017\)](#), contributions, even in a one-shot game, are not necessarily a pure reflection of own preferences for contributing, because they can also reflect beliefs about how much others will contribute.

⁴ The experimental sessions were conducted in person as students at the Burgundy School of Business were back to full in-person teaching by September 2021. We do not expect the sanitary context to affect our results.

same as before, with 3 subjects in each group. Subjects are informed that there are two changes compared to the first phase. First, the game will be repeated 10 times. Second, the composition of the group may be different compared to the first phase. Specifically, subjects are told that groups of three will be composed according to one of the following options:

- A) Subjects express a preference among possible future partners by ranking them. If this option is implemented, groups are formed according to subjects' preferences.
- B) The computer creates groups following a pre-defined procedure.

It is indicated in the instructions that subjects will be informed about the option that will be implemented before they start playing the first round of the second phase. However, before one of the two options is chosen to be implemented, subjects will be asked to express their preferences regarding future partners via a ranking, which will be applied if option A) is the one implemented. In other words, we asked all subjects to do the ranking. For some subjects, the computer sorted out subjects into groups exogenously according to either of two pre-defined rules that we present below. For other subjects, the computer applied the results from the ranking stage, thus implementing an endogenous sorting. Subjects were informed that groups will be formed according to their preferences expressed via the ranking with some probability less than one. We asked everyone to do the ranking instead of asking only subjects in the endogenous sorting condition because we wanted all our subjects to be aware of the alternative sorting arrangements. Our design choice is motivated by the fact that an individual may react differently to dictatorship if she lived her entire life under this type of regime and was never made aware that other political arrangements are possible than if she learns about regimes where voting exists. Our design choice is also motivated by Dal Bo et al.'s (2010) study where all subjects learned about the possibility of a fine but only some groups' votes were used to determine whether they played under the fine or not.

Two other design choices need to be discussed: (a) the ranking and endogenous grouping procedures, and (b) the details of exogenous sorting. We first present the ranking and endogenous grouping procedures. We used a similar procedure to Page et al. (2005). Each subject was shown a list, without other identifying information and in a random order, of each of the other 14 subjects' individual contribution to the group account in the first phase of the experiment (i.e., in the one-shot game).⁵ Unlike Page et al. (2005), ranking in our experiment was free. This way, we ensure that every subject expresses their preferences among all possible future partners. At the ranking stage, each subject has to type a number in a box next to the information about the other subject's contribution to the group account. Potential ranking numbers ran from 1 to 14, with 1 standing for the most preferred prospective partner. Ties are not allowed, meaning that the same ranking number cannot be assigned to two or more subjects. When all subjects had completed this process, the computer assigned subjects to groups by searching, first, for that group of three subjects the sum of whose mutual ranks of one another was the lowest among the universe of potential groups, then

repeating this process over the remaining subjects, to form the second and third groups, leaving the last three subjects in the fifth group. We used the Page et al. (2005) procedure with two considerations in mind: first, that their mechanism had in practice perfectly sorted the groups in the order of their levels of contributions to the public good⁶; second, that the mechanism's principle is relatively easy to understand for the subjects.

The second design feature that needs to be discussed is related to the exogenous sorting. Subjects know that groups can be formed either endogenously (based on the results of the ranking phase) or exogenously, in which case the computer forms groups following a pre-defined rule. However, subjects did not receive information about what exogenous sorting rules might be used. The only information we wanted them to have is that groups are either the result of their choices or the outcome of an external actor's choices (the experimenter). This way, we can identify the effect of decision rights in forming groups. The features of the exogenously formed groups were left for the subjects to infer from the information they received about their new group's composition, there being no announcement to them about the principles behind the exogenous grouping. To make clear to the reader the distinction between the three conditions under which Phase 2 could be played, we present here the regrouping procedure in case the computer implemented an exogenous sorting mechanism. Specifically, the computer implemented one of the following regrouping mechanisms:

- i) As in Gächter and Thöni (2005), the highest three contributors ended up in the same group, followed by the next to highest, etc. This condition is referred to as the *exogenous homogeneous sorting* (EXO-H), because it creates groups that are internally as homogeneous in contribution tendency as possible. The sorting achieved should be nearly the same as that obtained using the endogenous sorting mechanism from Page et al. (2005), if endogenous sorting outcomes resemble those in our experiment.
- ii) Alternatively, the 15 subjects were ordered according to their contribution to the group account in the one-shot game, creating three categories: 5 subjects with highest contributions, 5 subjects with next to highest contributions, and 5 subjects with the lowest contributions in the one-shot game. Using this initial division, well-mixed or diverse groups of three were formed by assigning to each group one subject from the 5 highest contributors, one subject from the intermediate category, and one subject from the 5 lowest contributors. This condition is referred to as the *exogenous mixed sorting* (EXO-M). We opted to form mixed groups instead of randomly created ones, as in the previous literature, because we wanted to control for the homogeneity of types. In randomly created groups, it may happen that, by chance, high contributors end up grouped together, intermediate types are in the same group, and low types interact with other low types. By creating mixed groups, we control for the group composition even in the control condition, contrary to other experiments where it is assumed that the type composition of groups in the baseline is mixed.⁷

⁵ It is worth noting that our design includes an unexpected use of elicited decision data. In particular, when subjects play the initial one round game, they are not informed that their decisions will play a role in the second part of the experiment, which they do by influencing the placement into groups either by computer algorithm or based on one another's preference rankings which can only be conditioned on first game contribution information. Because group assignments for the second part can influence second part payoffs, this previously unannounced use of first game decisions can be said to amount to payoff consequences of the first game decisions about which players do not have information at the time of making those decisions. Such a design feature is not judged to be deceptive by, and is indeed used by, numerous economic experimentalists. For a review of what is considered deception in experimental economics, see Hertwig and Ortmann (2008) and Charness et al. (2022).

⁶ While the procedure is not in principle strategy-proof, subjects tend to assign more favored ranks to those who contributed more, ignoring the potential advantage of other strategies, and the emerging groups thus perfectly reflected the order of contribution levels. Note well the role of mutual preference, preventing low contributors from simply selecting themselves into groups of high contributors. A variant of the procedure has similar effect in Kamei and Putterman (2017).

⁷ Ones and Putterman (2007) is the only other experiment of which we are aware that set up groups to be intendedly well mixed for their first five periods (in a 25 period experiment), in order that subjects' initial behaviors, that were used later to sort them with those who behaved similarly, are not attributable to what behaviors they happened to randomly meet.

After subjects completed the ranking stage, the computer selected one option among options A and B, chose between i) (EXO—H) and ii) (EXO-M) if B was selected, and formed groups for Phase 2 accordingly. Subjects were informed of the option (A or B) selected, and were informed about the individual contribution of each of their new partners in the one-shot game of Phase 1. We informed subjects about the individual contributions of their new partners in the previous phase (Phase 1 one-shot game) to ensure that subjects in all conditions receive information about the types of their new group mates. Given the amount of information provided to subjects in Phase 2 and the existence of the ranking stage, we checked subjects' comprehension of Phase 2 instructions using a set of comprehension questions. All subjects correctly answered the comprehension questions (see Instructions in the Online Material for phase 1 and 2 instructions).

We have 90 subjects who participated in the ENDO condition, 90 subjects who participated in the EXO—H condition, and 90 subjects who participated in the EXO-M condition. Given that we implemented groups of 3, we have 30 groups in each condition.

4. Predictions

In this section, we present our predictions. A first straightforward prediction is that players with selfish money-maximizing preferences will contribute 0 in all conditions. In contrast to this prediction, however, public goods experiments employing parameters similar to our game have found that subjects contribute between 40 % and 60 % of their endowment in the first round and then reduce their contributions to approach the game theoretic prediction in the last round of the game (Gächter, 2014). We therefore expect subjects in our conditions to behave differently from the theoretical prediction of complete non-contribution, and we look for suggestions in past behavioral and experimental literature in order to anticipate differences within the generally observed pattern between our ENDO, EXO—H, and EXO-M sorting conditions.

After first discussing predictions regarding differences in overall cooperation levels between our three conditions, we then explore potential differences between experimental conditions for individuals of different disposition, i.e. subjects with high cooperative dispositions, ii) subjects with intermediate cooperative dispositions, and iii) subjects with low cooperative dispositions.

The group formation literature presented in the previous section has shown that overall cooperation is significantly higher when groups are composed of like-minded individuals, whether they are sorted endogenously (e.g., Aimone et al., 2013; Brekke et al., 2011; Hauge et al., 2018; Page et al., 2005) or exogenously (Burlando and Guala, 2005; Gächter and Thöni, 2005; Gunthorsdottir et al., 2007; Kimbrough and Vostroknutov, 2016; Ones and Putterman, 2007), than when groups are randomly created. The existing literature on group formation in public goods games compares an endogenous or exogenous sorting mechanism to a baseline condition in which groups are assumed to be composed of different types (mixed groups). In our experiment, we control for the composition of groups in all conditions, including the control one (EXO-M). The mechanism that is put forward in the group formation literature to explain higher cooperation in endogenous and exogenous sorting conditions compared to mixed groups is that high levels of cooperation are achieved when conditional cooperators expect to interact with like-minded others. In the case of free-riders, they contribute at low levels whether they are in sorted or unsorted groups. Sorting subjects in like-minded groups, therefore, prevents cooperation from unraveling by screening and separating defectors from conditional cooperators.

Based on findings from the previous literature, we formulate our first prediction.

Prediction 1: Overall, we expect higher average rates of cooperation in the ENDO sorting and EXO—H sorting compared to the EXO-M sorting condition.

Our second prediction is related to the comparison between our ENDO and EXO—H sorting conditions. The experimental research that investigates the existence of an intrinsic value of decision rights (e.g., Bartling et al., 2014) seems to suggest that cooperation should be higher in ENDO than in EXO—H. However, none of the studies discussed has choice of group members as the outcome that is determined either endogenously by the subjects or exogenously by the experimenter, and while the endogenous group formation process gives subjects a voice in who they play with, there is no “majority rule” *per se*—indeed, only highest contributors are likely to end up with their most preferred potential partners, leaving most subjects to settle for those to whom they gave a less preferred rank. Given that the only existing evidence that can help us formulate our second hypothesis comes from the literature on an intrinsic value of decision rights, our prediction is as follows.

Prediction 2: Overall, we expect higher average rates of cooperation in ENDO than in EXO—H.⁸

Individuals may react differently to being assigned to one of our conditions depending on their cooperative type. Hence, we further discuss factors that may explain differences between our three sorting mechanisms by subjects' initial cooperation propensities.⁹ Using the same tripartite division that we apply to form mixed groups, we distinguish between high cooperators, intermediate cooperators, and low cooperators. Using the same classification, we obtain three types of subjects in each session, regardless of the experimental condition. Subjects may respond differently to their grouping condition not only due to their different initial propensities and beliefs about cooperation, but also because they are differently affected by grouping procedures, e.g. only high contributors are likely *ex post* to be grouped with their most preferred partners.

By hypothesis, in the ENDO condition, and by design, in EXO—H, subjects are grouped with like-minded others, while in the EXO-M condition, groups are mixed. Oliveira et al. (2015) found that one bad apple can spoil the barrel in a public goods game, supporting the conjecture of Fehr and Gächter (2000) that high contributors reduce their contributions in the presence of low contributors absent the option to reduce disadvantageous inequality by punishing. Following also the results of Gächter and Thöni (2005) and Gunthorsdottir et al. (2007), we expect that high cooperators will contribute more when grouped with like-minded others than when part of a mixed group.

Prediction 3: We expect high cooperators to contribute more in the experimental conditions where they are grouped with like-minded others (ENDO and EXO—H) than when playing in mixed groups (EXO-M).

Kreps et al. (1982) have shown that cooperation can emerge in a finitely repeated prisoner's dilemma if some individuals believe that others have non-selfish motivations. When playing in mixed groups which remain fixed for the duration of the game, low contributors may expect that the others in their group, who have contributed higher amounts to the group account in the one-shot game, may be willing to contribute more if they can observe some levels of cooperation from the beginning. By cooperating in the first rounds of the game and switching

⁸ Given that our second prediction relies on studies that investigate outcomes that are different from group cooperation, we decided to leverage the insights from future managers who may be confronted with the choice to let groups form endogenously or exogenously when it comes to the completion of joint tasks. We recruited 52 graduate students in management from Burgundy School of Business who played a simple guessing game. Management students were asked whether they expect average contributions under ENDO to be i) higher, ii) lower, or iii) the same as in the EXO-H sorting. Results from the guessing game show that 58.8% predicted higher average contributions in ENDO than in EXO-H, 23.5% predicted lower contributions in ENDO, and 17.6% predicted similar average contributions in the two conditions.

⁹ Contrary to predictions 1 and 2, the third prediction about the contributions by initial cooperation propensities was not pre-registered and, as such, is part of an exploratory analysis.

to free-riding in the last rounds, low cooperators may earn more than by free-riding from the beginning of the ten rounds. Such considerations are absent in the ENDO and EXO—H conditions where free-riders know that they are grouped with other free-riders. Low (Phase 1) contributors may also be conditional cooperators who contributed little due to pessimistic expectations about others, and when grouped with a middle and a high contributor, their contribution would accordingly rise.

Prediction 4: We expect low cooperators to contribute more in the EXO-M condition than in the ENDO and EXO—H conditions.

We cannot formulate any prediction regarding contributions of intermediate cooperators in the EXO-M condition compared to the ENDO and EXO—H sorting.

We have also not made a prediction regarding the effect of endogenous group formation, i.e. contributions under ENDO versus EXO—H, with respect to the individual contributor types. However, if Prediction 2 is correct, it implies that contributions will be higher in ENDO than in EXO—H for at least one category of contributors. A plausible conjecture is that the type most likely to manifest the predicted endogeneity effect is high contributors, because they are much more likely than others to find themselves in groups with the subjects they ranked as most preferred. We check for an endogeneity effect type by type when reporting our results, without raising the status of this conjecture to a numbered prediction.

5. Results

We start by providing a general overview of our results from Phase 1 where all subjects played the same one-shot public goods game. Subjects who were later allocated to the ENDO condition, contributed, on average, 9.28 tokens (s.d. 6.29) to the group account in Phase 1, while subjects who ended up in one of the exogenous sorting conditions contributed 10.92 (EXO—H, s.d. 6.91) and 10.72 (EXO-M, s.d. 6.61) tokens in Phase 1. However, average individual contributions are not significantly different across conditions in Phase 1 (ANOVA test, $p=.19$), which is what one should expect given that this phase is identical for all subjects.¹⁰ Thus, any differences in Phase 2 can be attributed to the type composition of groups and the procedure according to which like-minded subjects were grouped together. To make sure that any differences between ENDO and EXO—H can only be the result of procedural differences, holding constant the type composition, we now turn to an analysis of the type composition of groups.

5.1. Showing that the group composition is similar in ENDO and EXO-H

Before comparing cooperation levels in the ENDO and EXO—H conditions, we first show that our procedures to create groups endogenously and exogenously led to the formation of comparable internally homogeneous groups. We predicted that the Page et al. (2005) procedure to create groups endogenously would lead to the same group composition as the procedure used by Gächter and Thöni (2005) when they created groups exogenously: the highest contributors end up in the same group, the next to highest are grouped together, and the lowest contributors are in the same group. We now show that our endogenous group formation procedure was successful in mimicking what happened in EXO—H.

We classified subjects into three types based on their contributions in the one-shot game. We had 15 subjects in each session. The top five contributors are labeled as **high** type players. The next five contributors are labeled as **intermediate** type players. Finally, the lowest five contributors are labeled as the **low** type players. In case of a tie, the computer randomly broke the tie. Average contributions from Phase 1 show that our low types contributed 3.45 tokens, while intermediate types

and high types contributed 9.89, and 17.58, respectively.

Table 1 shows for each type of player the probability of having as future partners all other types. In the ENDO sorting condition, a high type has a probability of 84 % to have other high type players as Phase 2 partners (i.e., on average 84 % of other group members were high types). In the EXO—H condition, a high type has a probability of 87 % to have other high types as their partners. The difference is very small and statistically insignificant (2-sample test for equality of proportions, $p=.719$). For low type players, they have a probability of 80 % to meet like-minded partners in the ENDO sorting, and a 87 % chance to do so in the EXO—H sorting condition. The difference, although slightly higher than for high types, is still statistically insignificant ($p=.287$). For intermediate types, they have a 67 % chance to be grouped with like-minded others in the ENDO sorting, and a 73 % chance to be grouped with other intermediate type players in the EXO—H sorting. The difference is again statistically insignificant ($p=.474$). In the case of high and low types, the probability of meeting like-minded others is very high, ranging from 80 % to 87 %. It is lower in the case of intermediate types because in each session, we had 5 intermediate type players but groups of 3. Therefore, by design, two intermediate type players ended up grouped with other types: one intermediate type ended up in a group where the two others were high type players, while another intermediate type player was assigned to a group where the two others were of a low type.¹¹ It is worth noting that in the EXO—H sorting condition, the intermediate type player who ended up in a group where the two others were of a high type had a relatively high contribution compared to the other intermediate type players. That is, we followed the procedure from Gächter and Thöni (2005) and grouped together the first three contributors, then the next three ended up in the same group (two high type players and one intermediate type), until the lowest three were put in the same group (Fig. 1).

Thus, in EXO—H, subjects were ranked from the highest to the lowest contributor and people were grouped based on their contributions. This means that groups are homogeneous even when, according to our *post hoc* classification, we have groups composed of two high type players and one intermediate type. Furthermore, we can provide additional evidence that groups are also homogenous in the ENDO sorting condition. Fig. 2 shows the ranking patterns in all sessions. For each session, we show the relationship between the sum of ranks received and subjects' contributions in the one-shot game. According to our procedure, homogeneous groups will be formed if all subjects prefer to have cooperative partners. In terms of ranking, it means that the highest contributor in a session should have the lowest sum of ranks. Fig. 2 clearly shows that there is a strong correlation between subjects'

Table 1
Probabilities to meet other types conditional on own type.

Treatment	Pr(High)	Pr(Intermediate)	Pr(Low)
<i>High types</i>			
ENDO	0.84	0.14	0.01
EXO—H	0.87	0.13	0.00
EXO-M	0.33	0.33	0.33
<i>Intermediate types</i>			
ENDO	0.14	0.67	0.19
EXO—H	0.13	0.73	0.13
EXO-M	0.33	0.33	0.33
<i>Low types</i>			
ENDO	0.01	0.19	0.80
EXO—H	0.00	0.13	0.87
EXO-M	0.33	0.33	0.33

¹⁰ We also run multiple pairwise comparisons (with Bonferroni correction) finding no significant difference across any pair of conditions.

¹¹ Recall that the division into three “types,” which renders impossible perfect homogeneity of the exogenously formed homogeneous groups, is motivated by the goal of having perfectly mixed groups of three in the EXO-M condition.

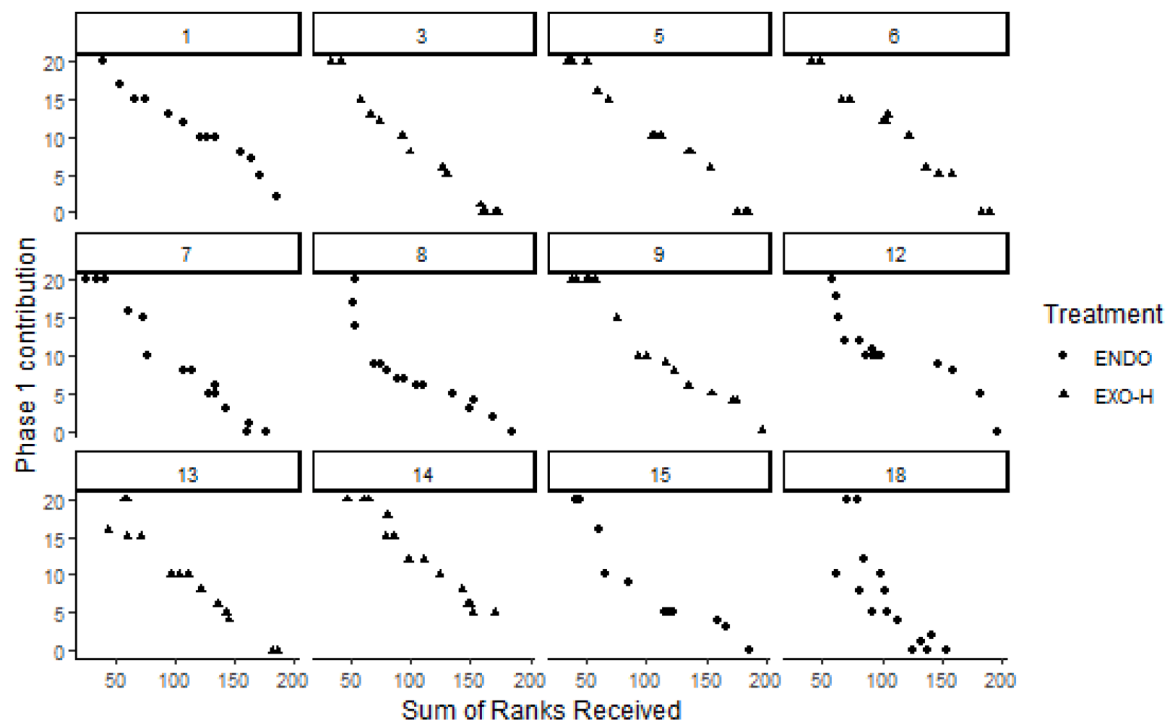


Fig. 1. Relation between ranks received and phase 1 contribution for each player across sessions of the ENDO and EXO—H treatments. Note: each point represents a participant in a given experimental session. Plot facets report session numbers.

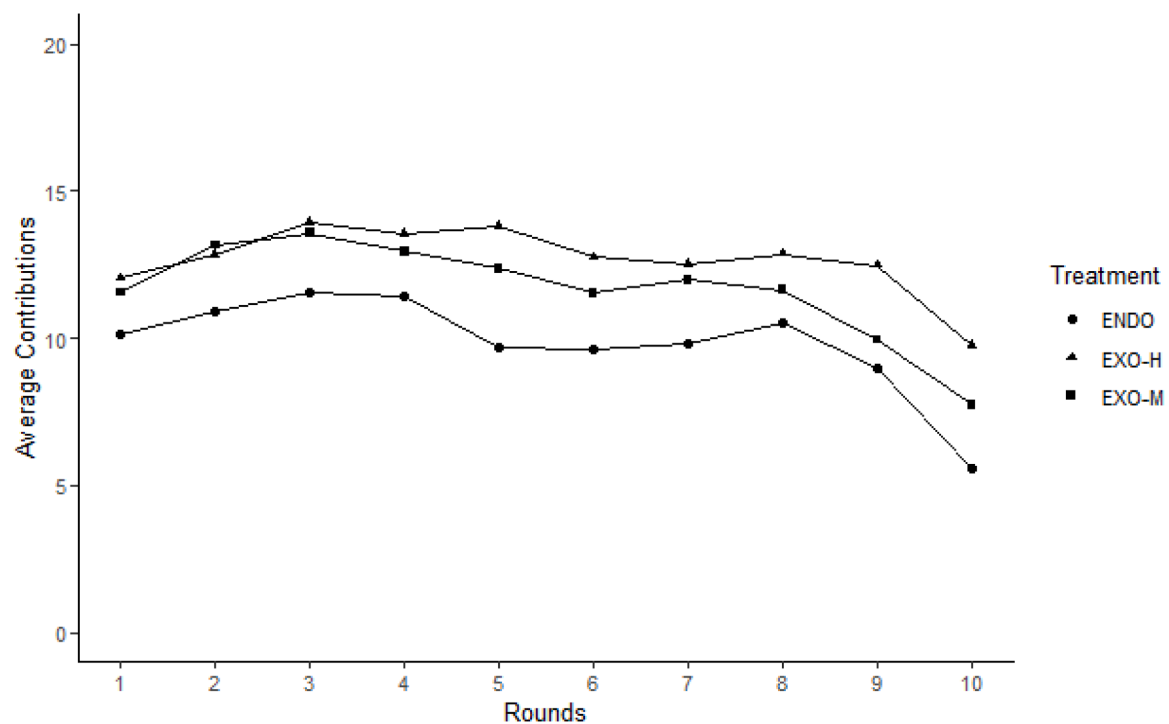


Fig. 2. Individual average contributions over rounds of Phase 2.

contributions and the sum of ranks received: at the ranking stage in our experiment, high contributors have a lower sum of ranks than low contributors.¹²

¹² Spearman coefficient of correlation ranging from 0.89 to 0.99, all p -values < .001

When comparing the final group composition and a participant's two most preferred partners, we find that, on average, 19 % of subjects end up with their top two preferred partners in both the ENDO and EXO—H conditions. Merging the two conditions (ENDO and EXO—H) and distinguishing between subject types, we find that 47 % of high cooperators end up with their two most preferred partners (22 % end up with only one most preferred partner); for the intermediate cooperators, 8 % end

up with two of their most preferred partners (18 % end up with only one); for the low cooperators, 3 % end up with their two most preferred partners (13 % end up with one). We notice that between the ENDO and EXO—H conditions, there is no statistical difference in the probability for high cooperators to end up with their two most preferred partners (Fisher's exact test, $p=.07$). Similarly, for intermediate cooperators, the probability for these players to be grouped with their two most preferred partners is not different between the ENDO and the EXO—H sorting ($p=.99$). The same applies to the low cooperators ($p=.49$). We therefore conclude that the type composition of our groups in the ENDO and EXO—H conditions is not different from each other.

5.2. Testing our predictions

Fig. 2 shows the evolution in the average individual contributions over the 10 rounds of the Phase 2 across our three conditions. Average contributions range from about half to two-thirds of the endowment and are generally stable until mild downward trends in the final periods. There is a clear difference in contributions between the ENDO condition and the other two exogenous conditions, EXO—H and EXO-M, starting from the first round of the second phase. Contrary to expectation, average contributions in ENDO are lower than in the other conditions and remain at lower levels consistently for all rounds.

We test our first prediction that average cooperation rates are higher in the ENDO and EXO—H sorting conditions than in EXO-M (our control condition). Table 2 shows the individual contributions in Phase 2 averaged over the 10 rounds across conditions. It indicates that average cooperation rates in the ENDO sorting are lower (9.82 tokens) than in EXO-M (11.60 tokens), while average cooperation rates in EXO—H sorting (12.70 tokens) are higher than in EXO-M. First round average contributions are statistically indistinguishable across treatments (pairwise Mann-Whitney ranksum test, $p\text{-value} > 0.05$).¹³

While non-parametric tests show only a significant difference between ENDO and EXO—H, this approach does not account for data dependencies over rounds. To analyze contribution behavior at the individual level, we estimate OLS regression models with standard errors clustered at the group level. We also report models using random effects at the individual and group level in the Appendix (see Table S1).

Table 3 (models 1–2) reports results of the comparison between EXO-M and each of the other sorting mechanisms (ENDO and EXO—H). We find no statistically significant difference between EXO—H and EXO-M as well as between ENDO and EXO-M, a result that goes against our first prediction.¹⁴ Additionally, Table S2 from the appendix reports a decay in cooperation levels over time in both ENDO and EXO-M conditions, however it appears significantly less pronounced in the EXO—H

Table 2

Average contributions in phase 2 of the experiment (repeated PGG) across conditions.

Treatment	Avg. Contributions	SD	N. subjects
ENDO	9.82	5.55	90
EXO—H	12.70	5.99	90
EXO-M	11.60	5.06	90

¹³ All p -values are adjusted for multiple comparisons using Bonferroni method. We obtain similar results also with parametric tests. Following our pre-registration (<https://aspredicted.org/zi95w.pdf>), we also compare averages over all rounds using a pairwise Mann-Whitney ranksum test and find a significant difference only between ENDO and EXO-H conditions ($p\text{-value} < 0.01$). However, we do not find any significant difference when performing other comparisons between our experimental conditions.

¹⁴ The same results hold when we control for the phase 1 contributions in the one-shot game.

Table 3

OLS models of contributions.

Baseline treatment	Model 1 EXO-M	Model 2 EXO-M	Model 3 ENDO	Model 4 ENDO
(Intercept)	13.51*** (0.85)	24.40*** (7.67)	11.68*** (1.04)	22.84*** (7.84)
ENDO	−1.83 (1.26)	−1.56 (1.26)		
EXO—H	1.01 (1.32)	1.08 (1.33)	2.84* (1.40)	2.64 (1.38)
Round	−0.34*** (0.06)	−0.34*** (0.06)	−0.34*** (0.06)	−0.34*** (0.06)
EXO-M			1.83 (1.26)	1.56 (1.26)
Demographics	No	Yes	No	Yes
Num. Obs.	2700	2700	2700	2700
Num. participants	270	270	270	270
Num. groups	90	90	90	90

Notes: OLS models with errors clustered at the group level. ** $p < 0.01$.

*** $p < 0.001$.

* $p < 0.05$.

^ $p < 0.10$.

condition.

Result 1: Overall, we find no higher average rates of cooperation in the ENDO sorting and EXO—H sorting compared to the EXO-M sorting condition.

We now turn to the comparison between ENDO and EXO—H. Based on the literature finding a democracy premium and an intrinsic value of decision rights as well as on guesses made by the 52 graduate students in business who participated in our guessing game, we formulated our second prediction that we should observe higher average rates of cooperation in the ENDO sorting than in the EXO—H sorting condition. We report results in the right columns of Table 3 (models 3–4). Estimates show a significant difference between the ENDO and the EXO—H is of around 5 tokens, yet significance reduces to 10 % level when adding demographic controls. In both cases, the direction of the effect is opposite to the one predicted.

Result 2: Overall, we find higher average rates of cooperation in the EXO—H sorting than in the ENDO sorting condition.

Given that individuals may react differently to being assigned to one of our conditions depending on their cooperative type, we further provide evidence on how different types behave across conditions. In the ENDO and EXO—H sorting conditions, subjects are grouped with like-minded others, while in EXO-M, groups are mixed. Given the results from Oliveira et al. (2015) who found that one bad apple can spoil the barrel, we predicted that high cooperators will contribute more when grouped with like-minded others than when being part of a mixed group. As we showed that high cooperators are grouped with other high cooperators both in ENDO and EXO—H, our third prediction stated that high cooperators should contribute more in ENDO and EXO—H than in EXO-M sorting condition.

Fig. 3 shows the evolution in average individual contributions over the 10 rounds of the second phase across the three conditions and by type. For high cooperator types, there is a substantial difference in cooperation levels between EXO—H and EXO-M sorting. It appears that high cooperators contributed more in EXO—H than in EXO-M over the 10 rounds. But when comparing high cooperators' contributions between the ENDO sorting and EXO-M, it seems that the two have very similar cooperation levels over the 10 rounds. Regression results reported in Table 4 confirm the visual interpretation. High cooperators contribute significantly less in EXO-M than in EXO—H sorting (see model 1), while there is no statistically significant difference between ENDO and EXO-M. All results hold even when controlling for individual

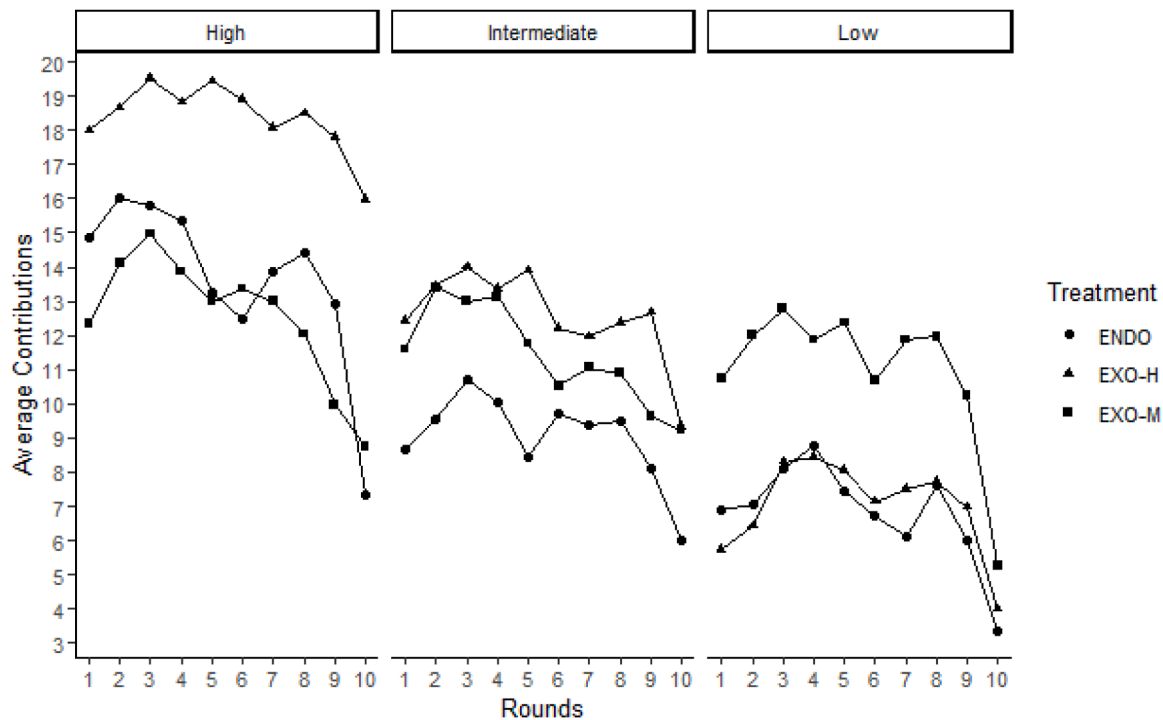


Fig. 3. Average individual contributions over periods by player types and treatments.

Table 4
OLS models of high types' contributions.

Baseline treatment	Model 1 EXO-M	Model 2 EXO-M	Model 3 ENDO	Model 4 ENDO
(Intercept)	14.90*** (1.10)	20.43 (12.49)	15.99*** (1.29)	21.73 (12.63)
ENDO	1.09 (1.62)	1.30 (1.61)		
EXO-H	5.83*** (1.25)	5.94*** (1.32)	4.74** (1.46)	4.64** (1.51)
Round	-0.43*** (0.09)	-0.43*** (0.09)	-0.43*** (0.09)	-0.43*** (0.09)
EXO-M			-1.09 (1.62)	-1.30 (1.61)
Demographics	No	Yes	No	Yes
Num. obs.	900	900	900	900
Num. participant	90	90	90	90
Num. groups	55	55	55	55

Notes: Standard errors clustered at the group level. * $p < 0.05$.

*** $p < 0.001$.

** $p < 0.01$.

characteristics (see models 2 and 4).¹⁵

Fig. 3 also shows that contributions are lower rather than higher in ENDO than in EXO-H, and it is apparent that this holds even for high cooperators, who were grouped with other high cooperators. The right columns of Table 4 (models 3–4) confirm that the difference is statistically significant.¹⁶

¹⁵ Overall, the data show a clear difference between contribution levels across types. However, one might expect that middle-type groups, formed from participants with intermediate contribution levels, would show greater within-group variation than high or low groups, which could lead to lower contributions. However, this is not supported by the data: variance in Phase 1 contributions is similar across types, with values of 3.34 for High types, 3.26 for Middle types, and 3.10 for Low types.

¹⁶ See Table S3 in the appendix for linear models with random effects at the individual and group level.

Result 3: High cooperators contribute significantly more in EXO-H than in EXO-M. There is no statistically significant difference between ENDO and EXO-M for high types. Contrary to our conjecture but in line with the failure of Prediction 2, high cooperators contribute less in ENDO than in EXO-H.

We now turn to our fourth and final prediction, which concerns the contributions of low cooperators. We followed the rationale from Kreps et al.'s (1982) model to form our fourth prediction that cooperating in the first rounds of the game and switching to free-riding in the last rounds can allow low types to elicit cooperation from the other types (intermediate and high type players in the same group). Such a strategy may prove more lucrative than if low types engaged in free-riding from the beginning of the 10 rounds. Such considerations are absent in ENDO and EXO-H conditions where free-riders know that they are grouped with other free-riders. From Fig. 3, it is clear that low cooperators contribute more in EXO-M than in ENDO and EXO-H sorting conditions. Prediction 4 is, therefore, the only one of our predictions that turns out to be in accord with subjects' decisions. This is confirmed by

Table 5
OLS models of low types' contributions.

Baseline treatment	Model 1 EXO-M	Model 2 EXO-M	Model 3 ENDO	Model 4 ENDO
(Intercept)	12.48*** (0.59)	8.13 (4.14)	8.32*** (0.59)	3.74 (4.35)
ENDO	-4.16*** (0.56)	-4.39*** (0.60)		
EXO-H	-3.94*** (0.56)	-4.03*** (0.58)	0.22 (0.56)	0.36 (0.57)
Round	-0.27*** (0.08)	-0.27*** (0.08)	-0.27*** (0.08)	-0.27*** (0.08)
EXO-M			4.16*** (0.56)	4.39*** (0.60)
Demographics	No	Yes	No	Yes
Num. obs.	900	900	900	900
Num. participant	90	90	90	90
Num. groups	56	56	56	56

Notes: Standard errors clustered at the group level. ** $p < 0.01$, * $p < 0.05$.

*** $p < 0.001$.

results presented in Table 5 showing significantly higher cooperation levels for low type players in EXO-M than in ENDO and EXO—H (see model 1).¹⁷ There is no difference in contribution levels among low types when comparing ENDO and EXO—H (see models 3–4). Furthermore, it appears visually that there is a much larger decline in contributions from round 8 to 10 by low types in EXO-M than in the other conditions. However, this is not statistically confirmed (ANOVA test, $p=.315$).

Result 4: We find that low cooperators contribute significantly more in EXO-M than in ENDO and EXO—H sorting conditions. The decrease in contributions over the last two rounds seems to be more dramatic in the EXO-M sorting condition compared to the other two conditions, suggesting that in EXO-M low types' higher contributions in the first eight rounds could be explained by their willingness to exploit the other types in their own group by temporary mimicking cooperators. However, this is not confirmed statistically.

Although we did not formulate any prediction regarding contributions of intermediate cooperators across our sorting conditions, it is worth noting that intermediate type cooperators do not contribute differently between EXO—H and EXO-M (see Table S5). This suggests that contribution behaviors of intermediate cooperators are not affected by whether they play with like-minded others or in more heterogeneous groups composed of a low and a high type. However, intermediate types contribute significantly more in EXO—H sorting than in ENDO and there is slightly higher cooperation from intermediate types in EXO-M than in ENDO, although the difference is not always statistically significant (see Table S5).

6. Discussion

Our main research question is whether individuals cooperate more when groups of like contributors are formed based on elicited preferences for partners or when such groups are formed exogenously by the experimenter. Our results indicate no statistically significant difference between each of our purposeful grouping conditions (EXO—H or ENDO) and our baseline (EXO-M). However, there seems to be significantly higher cooperation when individuals are placed in a group exogenously than when similar groups are formed endogenously. The result holds when distinguishing by types according to people's disposition to cooperate in a one-shot public goods game. We find that high and intermediate cooperators contribute significantly more when homogeneous groups are created exogenously than when such groups are formed endogenously.

Our research may help to shed light on which group formation mechanism can lead to higher cooperation in organizations. Furthermore, we contribute to two branches of the behavioral economics literature: a) the literature on cooperation in public goods games, and b) the literature showing that there is an intrinsic value to decision rights. We contribute to the cooperation literature by providing experimental evidence on the impact of endogenous and exogenous group formation, holding the type composition of groups constant. The case of high cooperators is particularly interesting because the high types were grouped with other high types in both ENDO and EXO—H, an outcome that corresponds to their elicited preferences in the ranking stage. For the high types, we can thus rule out the hypothesis that differences in cooperation levels can be the result of a form of frustration from not being matched with most preferred partners. We have also shown that observed differences in contributions of high types between ENDO and EXO—H cannot be attributed to differences in the type composition of groups. As noted in the results sections, there is no statistical difference in the probability for high cooperators to end up with their two preferred

partners between the ENDO and EXO—H sorting.

Ruling out frustration with the ranking outcome and differences in the type composition of groups as explanations for the differences in cooperation between high types in ENDO and EXO—H leaves us with an open question. We propose two possible mechanisms. The first mechanism that could explain our main result is based on high contributors' interpretation of the ranking outcome that was communicated to subjects solely in the ENDO condition. The ranking outcome in the ENDO condition may raise the salience of the view that others are self-interested, which may explain why high types contribute less in ENDO than in EXO—H starting from round 1 of phase 2. In effect, subjects receive information about others' ranking preferences in ENDO but not in EXO—H. Subjects might give thought to the reason that others give better ranks to higher contributors, and they may conclude that the reason is primarily material self-interest—each person benefits when they have other group members who contribute more. If the fact that fellow group members ended up in a specific group because they followed their self-interest in giving ranks is salient, then subjects may be more aware of the self-interest of the others in ENDO than in EXO, and it could lead them to make slightly lower predictions of how much the previously high contributors are going to contribute to the group account. Lower predictions of others' contributions could be self-fulfilling, and could be sufficient to explain why contributions are lower with endogenously formed homogeneous groups than with exogenously formed homogeneous ones. Unfortunately, we cannot test this conjecture because we did not elicit beliefs about how much others would contribute during phase 1 and phase 2 of the experiment.

The other possible mechanism that could explain the difference in cooperation between high types in ENDO and EXO—H is based on the concept of moral-licensing. Moral licensing is defined as “people's perception that they are permitted to take actions that could be seen as socially undesirable or morally questionable, due to history of moral behaviors” (Miller and Efron, 2010; p.118). The fact that a high contributor was highly ranked by others, which one could infer from the fact that s/he was able to get into a group with their most (or close to their most) preferred partners in the ENDO condition, may provide a license for subsequent bad deeds. Since the ranking outcome, potentially interpretable as a sign of moral approval by other “good actors,” was not communicated to subjects in the EXO—H condition, they received no such moral license, which could thus explain why high types contributed significantly less in ENDO than in EXO—H starting from round 1 of phase 2.

However, we cannot rule out the possibility of a spurious finding, especially given that the analyses that we performed at the type-level were not part of the pre-registration plan. In particular, the finding that high cooperators contribute more under EXO—H than under ENDO could be a one-off result, comparable to the early finding of Andreoni (1988) that stranger groups contribute more than partner groups, which subsequent studies did not replicate (Andreoni and Croson, 2008).

Another possibility, which we acknowledge as a plausible explanation for the failure of our predictions to hold, is that participants did not in practice experience a meaningful sense of empowerment from being assigned to the endogenous condition. While the ENDO mechanism is procedurally distinct from EXO in that it is based on participants' expressed preferences and is described as such in the instructions, the subjective experience of agency or control may have been limited. In light of prior research suggesting that perceived autonomy and participatory processes can influence motivation and behavior (Bartling et al., 2014; Deci and Ryan, 1985), it is conceivable that the procedural distinction between ENDO and EXO was insufficient to generate the psychological benefits typically associated with decision rights. This disconnect between the formal presence of choice and the felt experience of autonomy could help explain why cooperation did not increase under the endogenous condition.

Another limitation of our study is that while participants in all three treatments provided their full set of preferences over group members

¹⁷ We find similar results when estimating parameters using linear models with random effects at the individual and group level (see Table S4 in Appendix A).

before engaging in the main ten rounds of play, we did not undertake to elicit their preferences over the *method* of group formation. The fact that contribution levels were lower when groups were formed endogenously does not imply that most subjects would have expressed a preference for playing in exogenously formed groups, even if they had been informed of what exogenous group formation rule would be applied. It is not inconceivable that most participants would have expressed a preference that their say over potential group mates be counted. From a social welfare standpoint, any direct utility of having a say over interaction partners should not be assumed absent or ignored when comparing the desirability of different group formation methods.

7. Conclusion

Groups of people working on cooperative tasks are sometimes formed as the result of individuals' preferences to interact with like-minded others, but it may happen that the group composition is decided by some external authority (e.g., a manager). The experimental literature on group formation in social dilemma games has studied separately the effect of endogenous and exogenous mechanisms to create groups. Guido et al.'s (2019) survey and meta-analysis of the group formation literature provides only suggestive evidence on whether the two sorting mechanisms differ. While their paper suggests that cooperation levels are similar between endogenously and exogenously created homogeneous groups, their findings are only suggestive in that differences between experimental designs made it difficult to recreate the type composition of groups in the experiments included in their meta-analysis.

We designed a new experiment to study the effect of endogenously and exogenously formed homogeneous groups. Furthermore, our experiment allowed us to identify how individuals with different dispositions to cooperate (high, medium, low) behave in groups formed endogenously versus exogenously. The results were contrary to most of our predictions. Our main prediction was that average cooperation rates will be higher in endogenously formed groups than in exogenously formed homogeneous groups. A secondary prediction was that cooperation in the endogenously formed groups and in the exogenously formed homogeneous groups will be higher than in the baseline with mixed groups. The difference is, however, not statistically significant. An additional, more exploratory prediction, was that individuals with a high disposition to cooperate, as measured in a one-shot public goods game, will be more cooperative when being paired endogenously with like-minded others than when such groups are determined exogenously. Our predictions were based on the previous literature, and in particular on studies showing that there is an intrinsic value to decision rights and procedural autonomy (e.g., Bartling et al., 2014; Dal Bo et al., 2010; Deci and Ryan, 1985). Our findings contradict both of our predictions. We found that subjects' cooperation levels were significantly lower in endogenously formed groups than in exogenously formed homogeneous groups. This result holds especially for high cooperators, who contributed significantly less in endogenously formed groups than in exogenously formed homogeneous groups. However, this could be a one-off finding, comparable to the Andreoni (1988) result that stranger groups cooperate more than partner groups.

The other surprising result concerns the comparison between ENDO and EXO-M. We had predicted higher levels of cooperation in ENDO than in EXO-M, in particular for high cooperators who interact with like-minded others in ENDO but not in EXO-M. Our results indicate no statistically significant differences in the contribution behaviors of high types between the two conditions. However, it is worth noting that contributions of high types are if anything slightly higher in ENDO than in EXO-M in seven out of the ten rounds, but not statistically significantly higher. We also checked the average contributions of high types in the one-shot game of Phase 1 between ENDO and EXO-M to rule out pre-treatment differences and found no significant difference (16.23 and 17.73, respectively, in ENDO and EXO-M).

Another question raised by the comparison between ENDO and EXO-M as well as between EXO-H and EXO-M is whether our results call into question previous findings that there is higher cooperation when grouping subjects with like contributions than randomly allocating them to groups in public goods games (summarized in Guido et al., 2019). It is worth noting here that unlike previous studies, our EXO-M condition did not implement a random sorting. The fact that there are no statistically significant differences in overall cooperation rates and in those of the high cooperators, between ENDO and EXO-M, may be explained by the fact that low cooperators in EXO-M contributed more than low cooperators in ENDO or EXO-H. One reason for this difference may be that low cooperators in EXO-M, knowing themselves to be in a well-mixed group, had strategic reasons to contribute at relatively high levels. Thus, it is important to note that there are major differences between our design and the design of previous experiments that found higher cooperation in groups composed of like-minded subjects than in randomly formed ones. Our focus is on comparing exogenously with endogenously formed homogeneous groups, and we do not view our result as a strong rejection of the idea that interaction with similar types tends to slow the decay of cooperation in public goods games, although it raises the possibility that the strength of that effect is less robust than had been inferred from past research.

Our study provides insights that can be useful to researchers studying how to maintain high levels of cooperation in nonmarket interactions as well as when it comes to the design of teams that will have to work on joint tasks where some form of free-riding is possible. The question of how to form groups to allow for productive interactions between group members is relevant to many fields outside of academia. There are many examples of groups or associations that are formed for a limited time and then are reconstituted. Examples include law partnerships, academic departments, and athletic teams. Our results suggest that to achieve higher cooperation between group members, it is sometimes better to have an external authority or mechanism determine group composition than to employ a procedure based on individuals' preferences. But the extent to which our experimental findings should provide guidance in designing institutions, and which institutions they should be applied to, requires complementary study in relevant real-world settings.

CRedit authorship contribution statement

Andrea Guido: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Louis Putterman:** Writing – original draft, Supervision, Investigation, Conceptualization. **Rustam Romaniuc:** Writing – original draft, Validation, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.socrec.2025.102451](https://doi.org/10.1016/j.socrec.2025.102451).

Data availability

The data is publicly available in our researchbox.

References

- Aimone, J. A., Iannaccone, L. R., Makowsky, M. D., & Rubin, J. (2013). Endogenous group formation via unproductive costs. *Review of Economic Studies*, 80(4), 1215–1236.
- Andreoni, J. (1988). Why free ride? Strategies and learning in public goods experiments. *Journal of Public Economics*, 37, 291–304.
- Andreoni, J. (1995). Cooperation in public goods experiments: Kindness or confusion? *American Economic Review*, 85(4), 891–904.
- Andreoni, J., & Croson, R. (2008). Partners versus strangers: Random rematching in public goods experiments. In C. R. Plott, & V. L. Smith (Eds.), *Handbook of experimental economics results* (pp. 776–783). Elsevier. Vol. 1.

- Balliet, D. (2010). Communication and cooperation in social dilemmas: A meta-analytic review. *Journal of Conflict Resolution*, 54(1), 39–57.
- Bartling, B., Fehr, E., & Herz, H. (2014). The intrinsic value of decision rights. *Econometrica*, 82(6), 2005–2039.
- Bochet, O., Page, T., & Putterman, L. (2006). Communication and punishment in voluntary contribution experiments. *Journal of Economic Behavior and Organization*, 60, 11–26.
- Bowles, S., & Gintis, H. (2011). *A cooperative species*. Princeton University Press.
- Brekke, K. A., Hauge, K. E., Lind, J. T., & Nyborg, K. (2011). Playing with the good guys: A public good game with endogenous group formation. *Journal of Public Economics*, 95(9–10), 1111–1118.
- Burlando, R. M., & Guala, F. (2005). Heterogeneous agents in public goods experiments. *Experimental Economics*, 8, 35–54.
- Charness, G., Samek, A., & van de Ven, J. (2022). What is considered deception in experimental economics? *Experimental Economics*, 25, 385–412.
- Chaudhuri, A. (2011). Sustaining cooperation in laboratory public goods experiments: A selective survey of the literature. *Experimental Economics*, 14(1), 47–83.
- Dal Bo, P., Foster, A., & Putterman, L. (2010). Institutions and behavior: Experimental evidence on the effect of democracy. *American Economic Review*, 100, 2205–2229.
- Deci, E. L., & Ryan, R. M. (1985). *Intrinsic motivation and self-determination in human behavior*. Berlin: Springer Science & Business Media.
- Fehr, E., & Gächter, S. (2000). Cooperation and punishment in public goods experiments. *American Economic Review*, 90(4), 980–994.
- Festré, A., & Garrouste, P. (2015). Theory and evidence in psychology and economics about motivation crowding out: A possible convergence? *Journal of Economic Surveys*, 29(2), 339–356.
- Frey, B. S., & Jegen, R. (2001). Motivation crowding theory. *Journal of Economic Surveys*, 15(5), 589–611.
- Frey, B. S., Benz, M., & Stutzer, A. (2004). Introducing procedural utility: Not only what, but also how matters. *Journal of Institutional and Theoretical Economics*, 160(3), 377–401.
- Gächter, S., Kölle, F., & Quercia, S. (2017). Reciprocity and the tragedies of maintaining and providing the commons. *Nature Human Behavior*, 1(9), 650–656.
- Gächter, S. (2014). Human prosocial motivation and the maintenance of social order. *The oxford handbook of behavioral economics and the law*, chapter 2 (pp. 28–60). Oxford University Press.
- Gächter, S., & Herrmann, B. (2009). Reciprocity, culture and human cooperation: Previous insights and a new cross-cultural experiment. *Philosophical Transactions of the Royal Society B*, 364(1518).
- Gächter, S., & Thöni, C. (2005). Social learning and voluntary cooperation among like-minded people. *Journal of the European Economic Association*, 3, 303–314.
- Guido, A., Robbett, A., & Romaniuc, R. (2019). Group formation and cooperation in social dilemmas: A survey and meta-analytic evidence. *Journal of Economic Behavior and Organization*, 159, 192–209.
- Gunnthorsdottir, A., Houser, D., & McCabe, K. (2007). Disposition, history and contributions in public goods experiments. *Journal of Economic Behavior and Organization*, 62(2), 304–315.
- Hauge, K. E., Brekke, K. A., & Nyborg, K. (2018). Sustaining cooperation through self-sorting: The good, the bad, and the conditional. *Proceedings of the National Academy of Sciences*, 116(12), 5299–5304.
- Hertwig, R., & Ortmann, A. (2008). Deception in experiments: Revisiting the arguments in its defense. *Ethics and Behavior*, 18, 59–92.
- Kamei, K., & Putterman, L. (2017). Play it again: Partner choice, reputation building and learning in restarting, finitely-repeated dilemma games. *Economic Journal*, 127(602), 1069–1095.
- Kimbrough, E. O., & Vostroknutov, A. (2016). Norms make preferences social. *Journal of the European Economic Association*, 14(3), 608–638.
- Kreps, D. M., & Wilson, R. (1982). Reputation and imperfect information. *Journal of Economic Theory*, 27(2), 253–279.
- Markussen, T., Putterman, L., & Tyran, J.-R. (2014). Self-organization for collective action: An experimental study of voting on sanction regimes. *Review of Economic Studies*, 81, 301–324.
- Masclét, D., Noussair, C., Tucker, S., & Villeval, M.-C. (2003). Monetary and non-monetary punishment in the voluntary contributions mechanism. *American Economic Review*, 93(1), 366–380.
- Miller, D. T., & Effron, D. A. (2010). Psychological license: When it is needed and how it functions. *Advances in Experimental Social Psychology*, 43, 115–155.
- Oliveira, A., Croson, R., & Eckel, C. (2015). One bad apple? Heterogeneity and information in public good provision. *Experimental Economics*, 18(1), 116–135.
- Ones, U., & Putterman, L. (2007). The ecology of collective action: A public goods and sanctions experiment with controlled group formation. *Journal of Economic Behavior and Organization*, 62(4), 495–521.
- Ostrom, E., Walker, J., & Gardner, R. (1992). Covenants with and without a sword: Self-governance is possible. *American Political Science Review*, 86(2), 404–417.
- Page, T., Putterman, L., & Unel, B. (2005). Voluntary association in public goods experiments. *Economic Journal*, 106, 1032–1053.
- Raihani, N. (2021). *The social instinct: how cooperation shaped the world*. New York: St. Martin's Press.
- Rand, D. G., & Nowak, M. A. (2013). Human cooperation. *Trends in Cognitive Sciences*, 17, 413–425.
- Sutter, M., Haigner, S., & Kocher, M. G. (2010). Choosing the carrot or the stick? Endogenous institutional choice in social dilemma situations. *Review of Economic Studies*, 77, 1540–1566.
- Tirole, J. (2009). Intrinsic motivation, incentives, and social norms. *Revue Économique*, 60(3), 577–589.
- Tyran, J.-R., & Feld, L. L. (2006). Achieving compliance when legal sanctions are non-deterrent. *Scandinavian Journal of Economics*, 108(1), 135–156.